CS 374: Section A - Homework 8, Problem 1

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At any moment the card is in one of three **modes**: A (both stickers on), B (top sticker removed) or C (both stickers removed). The mode only moves forward, $A \to B \to C$, and a sticker may be removed at any station. While the card is in mode X, an edge e costs

$$w_X(e) = \begin{cases} \$(e)/2 \text{ if } system(e) = X \\ \$(e) \quad \text{otherwise.} \end{cases}$$

A **plan** is a walk from s to t in G together with the stations where the stickers come off. Its cost is the sum of $w_X(e)$ over the edges e of the walk, where X is the mode while that edge is used.

Graph

We build a directed graph $G' = (V', E')$ with three copies of G, one for each mode:

- **Vertices:** $V' = V \times \{A, B, C\}$. Vertex (v, X) means "we are at station v and the card is in mode X." So $|V'| = 3\ |V|$.
- **Travel edges:** For every edge $e = u \to v$ in E and every mode $X \in \{A, B, C\}$, G' has an edge $(u, X) \to (v, X)$ with weight $w_X(e)$. This is riding connection e without touching the stickers.
- **Sticker edges:** For every station $v \in V$, G' has an edge $(v, A) \to (v, B)$ and an edge $(v, B) \to (v, C)$, both with weight 0. These are removing the top sticker and the bottom sticker at station v. There are no edges back to an earlier mode, because a sticker cannot be put back on.

So $|E'| = 3\ |E| + 2\ |V|$. Every weight is $\$(e)$, $\$(e)/2$ or 0, so all weights are non-negative.

Algorithm

> Run Dijkstra's algorithm on G' from (s, A) and return $\min\{dist(t, A), dist(t, B), dist(t, C)\}$.

We start at (s, A) because both stickers are on when we leave the friend's house. We may arrive at the Transit Authority in any mode, so we take the best of the three copies of t. If none of them is reachable, all three distances are ∞ and there is no way to reach t.

Correctness

Every plan gives a path in G' with the same cost: Follow the plan step by step, starting at (s, A). When the plan rides an edge $e = u \to v$ in mode X, take the travel edge $(u, X) \to (v, X)$,

which has weight $w_X(e)$, the price of that ride. When the plan removes a sticker at station v, take the sticker edge $(v, A) \to (v, B)$ or $(v, B) \to (v, C)$, which has weight 0. The plan ends at t in some mode X, so this is a path from (s, A) to (t, X) whose weight equals the cost of the plan.

Every path in G' gives a plan with the same cost: Take a path from (s, A) to (t, X). Each travel edge $(u, Y) \to (v, Y)$ comes from an edge $e = u \to v$ of G, and riding e in mode Y costs $w_Y(e)$. Each sticker edge stays at the same station and moves the mode one step forward, so it is a legal sticker removal that costs nothing. The modes along the path only go forward, A then B then C, so the stickers come off in the allowed order. Reading the stations in order gives a walk from s to t in G, and the cost of this plan equals the weight of the path.

Therefore the cheapest plan costs exactly the smallest distance from (s, A) to one of (t, A), (t, B), (t, C), which is what the algorithm returns. Dijkstra's algorithm is correct here because every weight in G' is non-negative.

Example: In the handout's graph, the plan given there is the path

$$(s, A) \xrightarrow{1} (b, A) \xrightarrow{4.5} (e, A) \xrightarrow{0} (e, B) \xrightarrow{5} (d, B) \xrightarrow{3} (c, B) \xrightarrow{4} (f, B) \xrightarrow{0} (f, C) \xrightarrow{4} (t, C),$$

with total weight $1 + 4.5 + 5 + 3 + 4 + 4 = 21.5$, the value in the handout.

Running time

Building G' takes $O(|V'| + |E'|) = O(|V| + |E|)$ time: each edge of G gives three travel edges and each station gives two sticker edges, each with a weight found in $O(1)$ time. Dijkstra's algorithm on G' takes $O(|E'| \log|V'|) = O((|E| + |V|) \log|V|)$ time, since $\log|V'| = \log 3 + \log|V|$. Taking the minimum of three distances is $O(1)$. The algorithm runs in

$$\boxed{O((|V| + |E|) \log|V|) \text{ time.}}$$

Sources and collaborators: None.

LLM use: I used Claude (Anthropic) as a tutor for this problem: it helped me understand the problem and guided me, through explanation and questions, toward a correct approach. I wrote and typeset this entire submission myself.